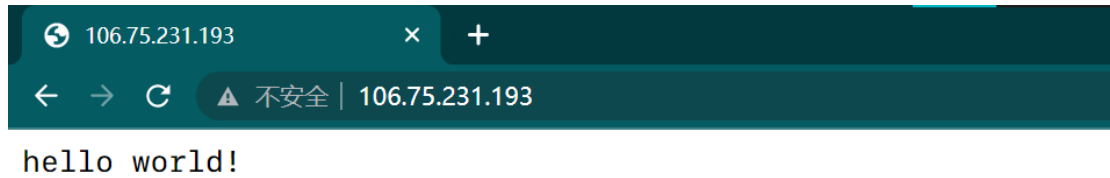


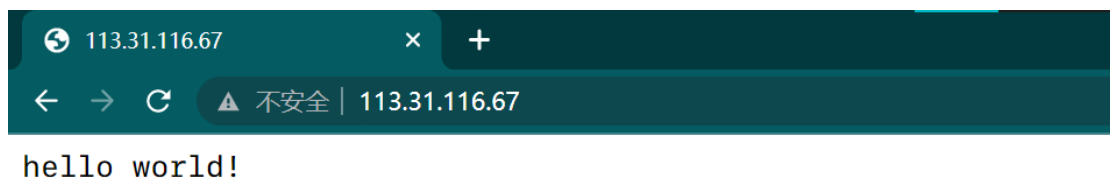
云计算实验六报告

作业 1:

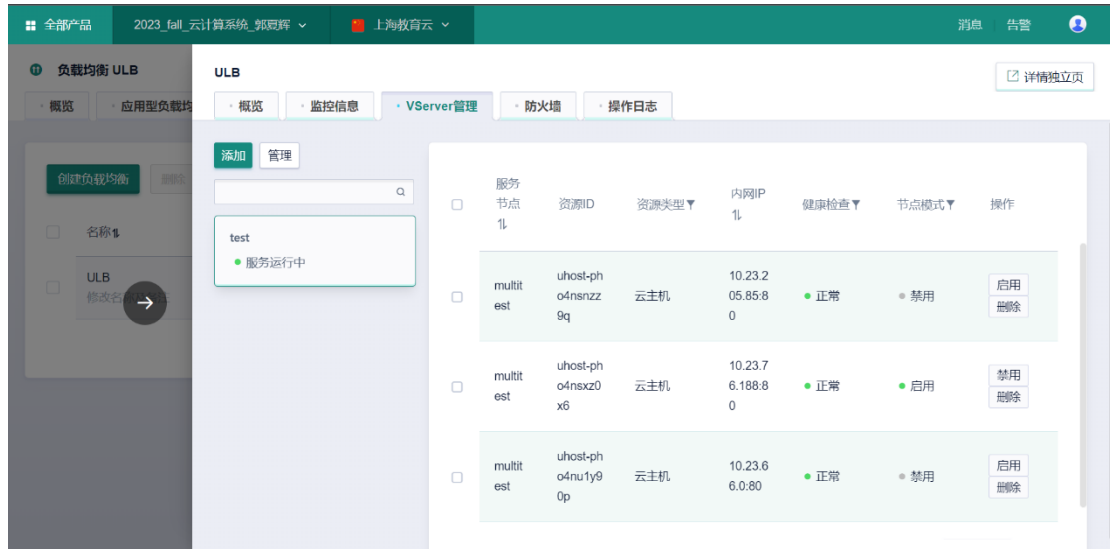
在打开负载均衡之前（云主机的外部 ip）:



打开负载均衡之后(负载均衡器的外部 ip):



作业 2:



作业 3:

启用一个结点:

```
Server Software: Werkzeug/3.0.1
Server Hostname: 113.31.116.67
Server Port: 80

Document Path: /
Document Length: 12 bytes

Concurrency Level: 1000
Time taken for tests: 27.827 seconds
Complete requests: 10000
Failed requests: 36
  (Connect: 0, Receive: 0, Length: 36, Exceptions: 0)
Non-2xx responses: 36
Total transferred: 1852412 bytes
HTML transferred: 123960 bytes
Requests per second: 359.37 [# /sec] (mean)
Time per request: 2782.680 [ms] (mean)
Time per request: 2.783 [ms] (mean, across all concurrent requests)
Transfer rate: 65.01 [Kbytes/sec] received

Connection Times (ms)
      min  mean[+/-sd] median  max
Connect:    3    5  4.2      4    24
Processing: 17 1313 4406.2   107 27782
Waiting:    10 1312 4406.3   106 27782
Total:      38 1318 4409.0   110 27794

Percentage of the requests served within a certain time (ms)
 50%    110
 66%    112
 75%    113
 80%    114
 90%   1233
 95%   13398
 98%   14480
 99%   27727
100%  27794 (longest request)
[root@10-23-68-101 ~]#
```

启用两个结点：

```
Server Software: Werkzeug/3.0.1
Server Hostname: 113.31.116.67
Server Port: 80

Document Path: /
Document Length: 12 bytes

Concurrency Level: 1000
Time taken for tests: 13.785 seconds
Complete requests: 10000
Failed requests: 0
Total transferred: 1850000 bytes
HTML transferred: 120000 bytes
Requests per second: 725.43 [#/sec] (mean)
Time per request: 1378.492 [ms] (mean)
Time per request: 1.378 [ms] (mean, across all concurrent requests)
Transfer rate: 131.06 [Kbytes/sec] received

Connection Times (ms)
          min  mean[+/-sd] median   max
Connect:    3      5   4.3      4     24
Processing: 11     534 1705.8    103  13740
Waiting:    10     532 1705.8    102  13738
Total:       36     539 1708.2    107  13756

Percentage of the requests served within a certain time (ms)
 50%    107
 66%    109
 75%    109
 80%    110
 90%    128
 95%   4378
 98%   6893
 99%   7626
100%  13756 (longest request)
[root@10-23-68-101 ~]#
```

启用三个结点：

```
Server Software: Werkzeug/3.0.1
Server Hostname: 113.31.116.67
Server Port: 80

Document Path: /
Document Length: 12 bytes

Concurrency Level: 1000
Time taken for tests: 4.479 seconds
Complete requests: 10000
Failed requests: 0
Total transferred: 1850000 bytes
HTML transferred: 120000 bytes
Requests per second: 2232.46 [#/sec] (mean)
Time per request: 447.936 [ms] (mean)
Time per request: 0.448 [ms] (mean, across all concurrent requests)
Transfer rate: 403.33 [Kbytes/sec] received

Connection Times (ms)
          min  mean[+/-sd] median   max
Connect:    3      5   4.7      4     26
Processing: 20     293  665.4     99  3595
Waiting:    9     292  665.4     98  3594
Total:       39     298  667.2    103  3612

Percentage of the requests served within a certain time (ms)
 50%    103
 66%    125
 75%    129
 80%    130
 90%    545
 95%   1761
 98%   3501
 99%   3540
100%   3612 (longest request)
[root@10-23-68-101 ~]#
```

观察可得，随着节点数量的增多，负载均衡的效果明显提升，time per request 下降十分明显（从 27.827 到 13.785 再到 4.479s）这里我感觉数据有点不正常，经过了多次测试 time per request 甚至出现了不减反增的情形（只有一个结点情况下甚至达到了 56s!），这应该就是网络因素的干扰吧。

作业 4:

(5×10^7) 请求 / 30 请求每秒每台 / 2 小时 / 3600 秒每小时 = 231.48 台

然后为了“度过峰值”，所以还要向上取整，即最少 232 台机器才可以平稳应对题述请求。